

Extracted parameters

Assumed WCET at the 95% empirical CDF point

Parameters extracted from 57 complete, unchanged full-v2 missions. The source campaign remains sealed; no new simulation was launched. All time values in this report and its new diagrams are seconds.

$F_i(c) = \text{count}(c_{i,k} \leq c) / N_i$; $C_i(\text{model}) = \inf\{c : F_i(c) \geq 0.95\} = \text{the } \text{ceil}(0.95 N_i)\text{-th ordered CPU sample.}$

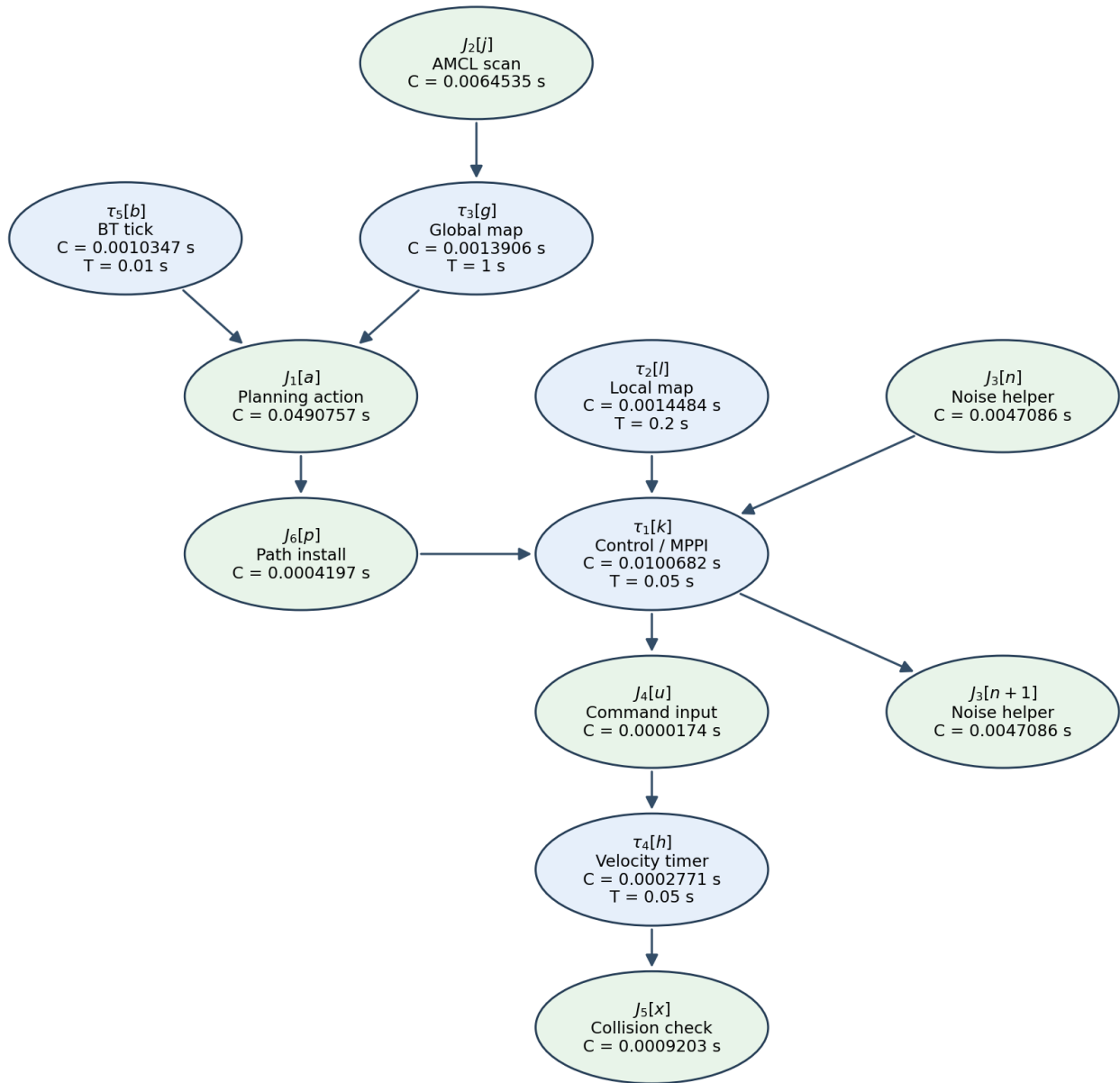
This model WCET is the user-selected CPU budget; it is not a certified worst-case bound. All new table/diagram time values are in seconds.

ID	Primary work unit	Type	C / assumed WCET (s)	Nominal T (s)	Mean interval (s)	Jobs
tau1	Control / MPPI	P	0.0100682	0.05	0.05000275	366220
tau2	Local map	P	0.0014484	0.2	0.20000543	91710
tau3	Global map	P	0.0013906	1	1.00000121	18345
tau4	Velocity timer	P	0.0002771	0.05	0.05000166	366870
tau5	BT tick	P	0.0010347	0.01	0.01001787	1828599
J1	Planning action	A	0.0490757	—	1.05355955	17418
J2	AMCL scan	A	0.0064535	—	0.23708611	77369
J3	Noise helper	A	0.0047086	—	0.05000273	366278
J4	Command input	A	0.0000174	—	0.05000205	366272
J5	Collision check	A	0.0009203	—	0.05000949	366193
J6	Path install	A	0.0004197	—	1.05342221	17418

For the five periodic named work units, $\text{sum}(C_i/T_i) = 0.319008600$. This accounts only for their selected CPU budgets; it excludes aperiodic work and other ROS/Gazebo/background demand and is not a schedulability result. The nominal hyperperiod is 1 s.

Book-style task/job scheduling DAG

Task/job DAG for the atomic result-delivery scheduling model
C: assumed CPU budget at 95% ECDF; T: nominal HOST period; all times in s

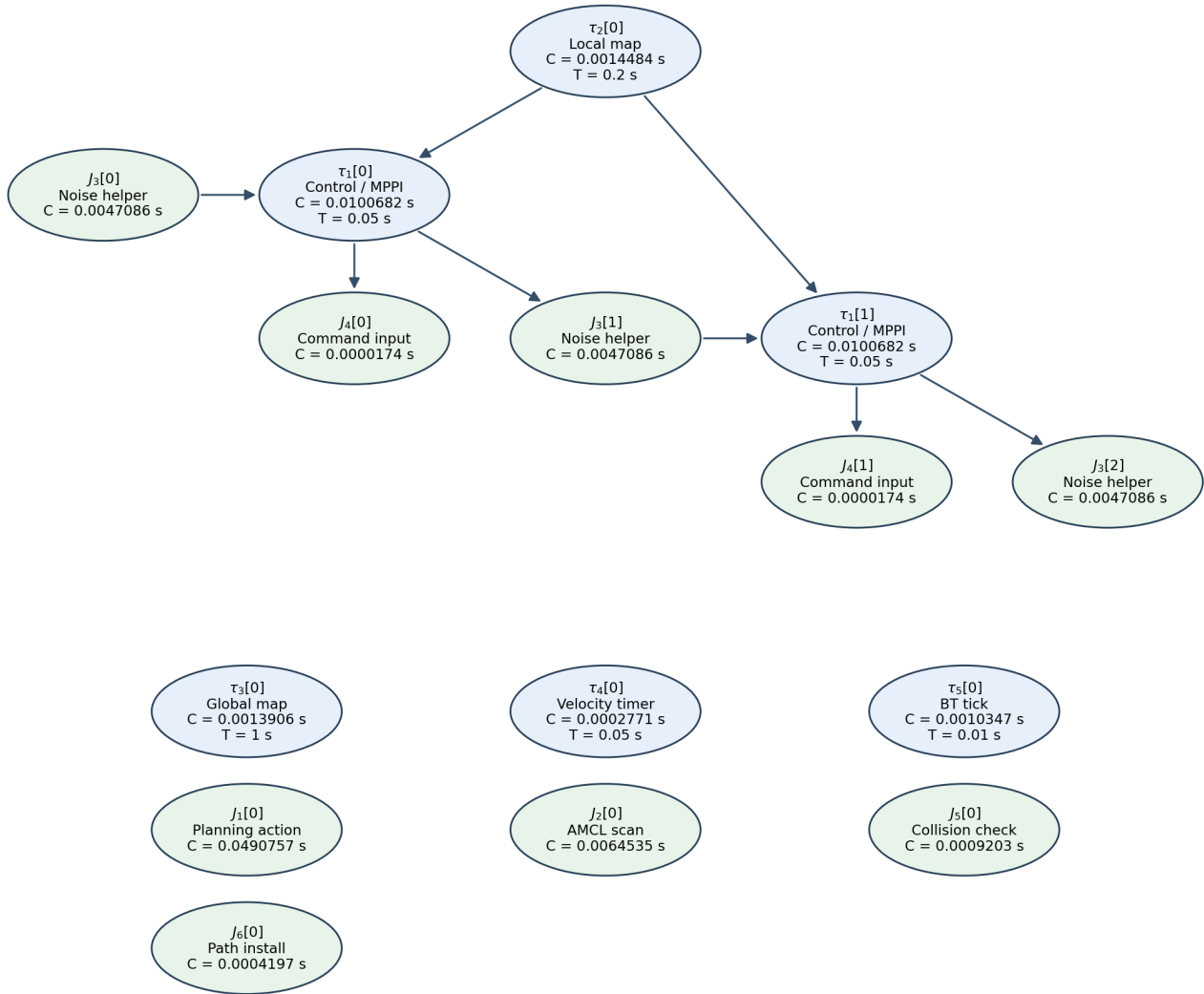


Arrows are selected data/trigger dependencies, not observed CPU execution order.
The TF edge is a source-backed model relation; exact producing TF job versions were not traced.
Cached-state indices select an available producer job; they do not force a fresh update every period.

Chosen scheduling abstraction: atomic output delivery at modeled job completion. Current ROS can publish/trigger before function return. This is not an extracted claim of whole-function completion precedence. Repeated helper jobs are indexed to unroll feedback.

Verified job dependencies

Measured job dependencies with CPU-order edges removed
C: assumed CPU budget at 95% ECDF; T: nominal HOST period; all times in s

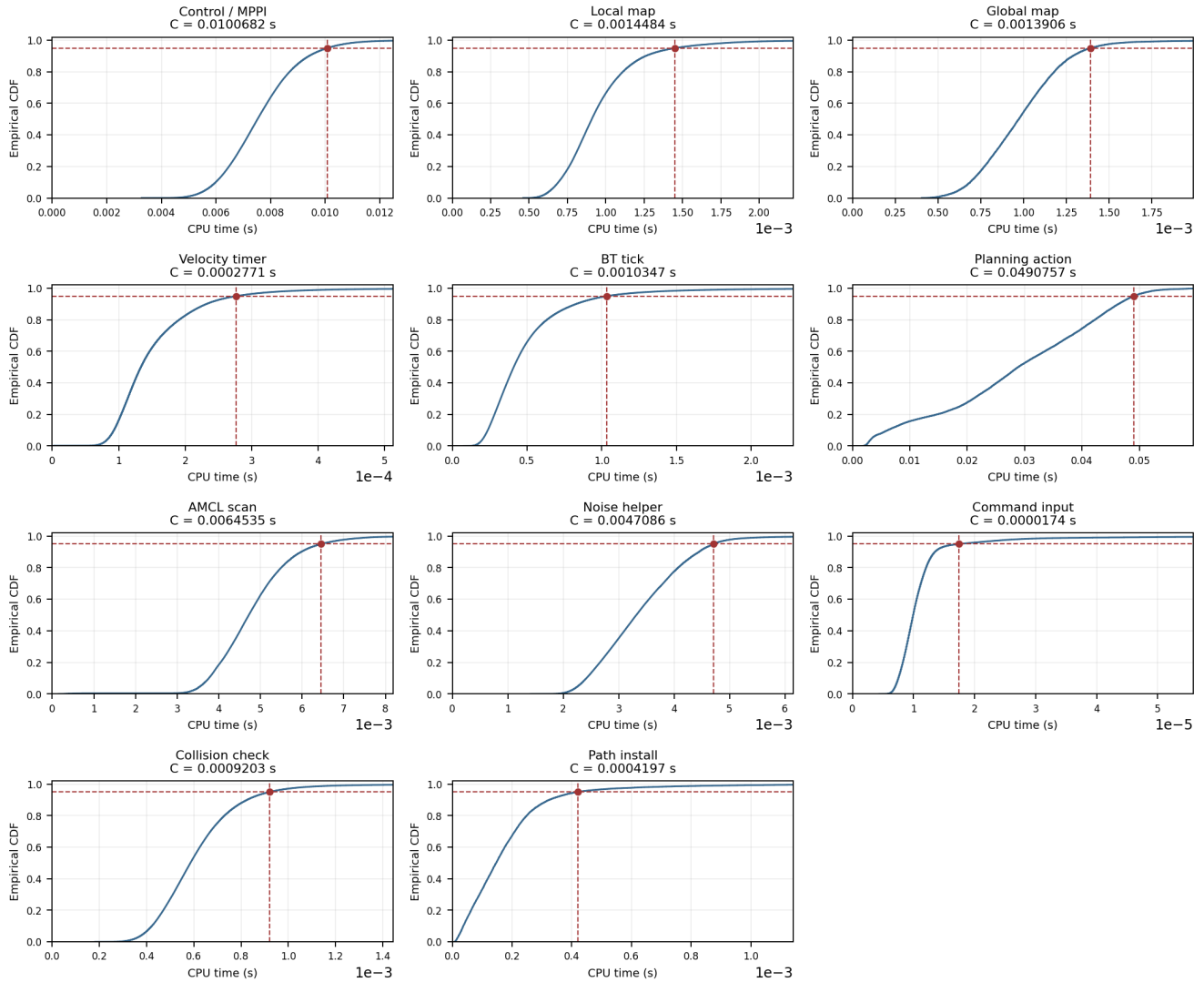


Bottom isolated vertices: other task representatives with no matched edge in this selected window.
Isolation does not establish global independence. Arrows identify observed producing/consuming phases;
atomic completion semantics apply only when the scheduling-model convention is adopted.

15 vertices, 8 dependency edges; 28 CPU-order edges removed. Other task representatives are isolated only within this selected window.

Empirical CPU CDFs

Exact 95%-ECDF CPU budgets from 57 complete missions
Curves subsampled for display; right tail view ends near Q99.5; quantiles use every job



The plotted curves are subsampled for readability. Every recorded job contributes to the exact nearest-rank quantile. Red lines mark the selected 95%-CDF budget.

Dependency meanings

Producer	Consumer	Selected data / trigger
amcl	global	selected map-to-odom TF state
bt	plan	action request
global	plan	selected committed global grid
plan	path	action path result
path	control	selected installed path
local	control	selected committed local grid
noise_old	control	selected committed noise
control	noise_next	noise generation trigger
control	command	command message
command	smooth	selected held command
smooth	collision	smoothed command message

Modeling decisions and scheduler readiness

C is the inclusive thread-CPU budget of the named primary work unit. T and observed inter-entry use HOST wall time. Both are expressed in seconds; identical units do not make their clocks interchangeable.

The user selects the 95%-ECDF point as the model WCET. It is an assumed simulation budget, not a certified bound on all executions. The CSV explicitly retains the fraction and count of jobs exceeding it.

For periodic implementations, use the configured nominal T. For aperiodic streams, T stays unassigned: their measured average inter-entry interval is descriptive and is not a guaranteed period or sporadic minimum. Trace arrivals can drive the initial scheduler.

The earlier report used NumPy linear-interpolation percentiles. This extraction instead computes the exact generalized inverse of the empirical CDF (nearest-rank order statistic). Both values are retained in the CSV to explain small numerical differences.

Whole-loop CPU budgets, when measured, are exported separately. Inclusive parent and nested-child costs must not both be charged. A parent/child call relation is containment; it is not an edge saying that the parent must finish before its child starts.

The book-style scheduling DAG adopts atomic output delivery at modeled completion. This matches a job-result delivery abstraction; current upstream ROS can publish/trigger before the enclosing function returns. The measured phase DAG remains the reference for those exact boundaries.

Cached-state edges select the already available producer version; they do not force each control job to wait for a fresh map, path or command update. Noise feedback is unrolled as old noise -> current control -> next noise.

The measured-job diagram removes all same-thread resource-order edges. An isolated representative means no exact dependency matched in this chosen trace window. It does not imply that its task family never interacts with other ROS components. No per-job TF version or exact global-grid producer is invented.

Task-family relations for planning, installed path and command smoothing follow the installed-version source/configuration. Their precise producer/consumer job selectors are model inputs, not new trace facts. The AMCL-to-global-map TF relation is source-backed at task-family level; exact TF job-version mapping and other live TF reads remain in the pass-through framework.

Scheduler development can start with these budgets, configured periodic releases and recorded aperiodic arrivals. Unlisted or unresolved infrastructure callbacks continue as pass-through background work in ROS. No invented C, T, deadline or priority is assigned to them, and they do not block the prototype.

Deadlines remain unassigned. No priority is inferred from historical execution order. Processor/core choices and dispatch policy belong to the next scheduling stage. Simulation was not rerun for this extraction.

Nested and whole-loop budgets, per-task sample counts, exceedances and prior linear Q95 values are available in the CSV/JSON artifacts. Original measurements and raw CSVs remain unchanged.

Reference: Buttazzo (2011), Section 2.2.2, printed pp. 28-29 / PDF pp. 45-46. Source/configuration evidence is preserved in the prior English measurement report.